E-Commerce Sales Intelligence

An Interactive Power BI Analytics Project

Microsoft Power BI Desktop · DAX · Python · Star Schema

Dataset period: January 2024 – December 2025

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Date: October 2026

Metric	Verified Value
Total Revenue	$1,888,956
Total Profit	$929,083
Profit Margin %	49.2%
Total Orders	8,456
Total Customers	2,047
Average Order Value	$223.39
Total Units Sold	17,840
Repeat Customer Rate	64.4%

All KPIs filtered to completed orders only. Source: FactSales.csv

1. Executive Summary

E-Commerce Sales Intelligence is a self-contained Power BI analytics project that transforms raw transactional, customer, and product data into a fully interactive five-page dashboard. It demonstrates end-to-end data analyst skills: schema design, data modelling, DAX measure authoring, RFM customer segmentation, and storytelling-grade dashboard design.

The project addresses a realistic e-commerce business problem: management needs a single pane of glass to monitor revenue and profitability, understand which products and customers drive value, diagnose geographic and channel performance, and surface actionable recommendations from a two-year sales history (Jan 2024-Dec 2025).

All headline figures in this report are independently verified against the source CSV files. Figures that could not be independently confirmed are labelled accordingly.

2. Business Problem and Objectives

A growing e-commerce retailer selling home goods across eight categories faces common analytical challenges. This project is designed to answer:

- Revenue health — Is the business growing? Where are the seasonal dips?
- Category and product profitability — Which categories and SKUs drive margin?
- Customer quality — How many customers return? Which RFM segments hold the most value?
- Geographic opportunity — Which US states dominate, and where is there upside?
- Channel efficiency — Which acquisition channels deliver the best revenue per customer?
- Order health — What revenue is lost to returns and cancellations?
- Discount risk — Do discounts expand volume enough to offset margin compression?

Intended Users

Executive leadership (Pages 1 and 5), marketing managers (Pages 3 and 4), product category managers (Page 2), and the data team (all pages plus source files).

3. Dataset Overview

All source data was created by the project author for this portfolio project. It contains no real personal or commercial data and is suitable for public portfolio publication.

File	Table	Rows	Description
FactSales.csv	FactSales	15,668	Order line items (all statuses)
DimCustomer.csv	DimCustomer	3,600	Customer dimension with RFM attributes
DimDate.csv	DimDate	731	Date dimension: Jan 2024 to Dec 2025
DimProduct.csv	DimProduct	90	Product master with cost and margin
CustomerRFM.csv	CustomerRFM	2,047	Pre-computed RFM scores and segments
BusinessInsights.csv	BusinessInsights	8	Narrative insight reference table

Order Status Distribution

Status	Rows	% of Total	Used in KPIs?
Completed	14,339	91.5%	Yes — all revenue KPIs
Returned	768	4.9%	Return metrics only
Cancelled	561	3.6%	Cancellation count only

Data quality: No null values detected in key join columns. DimCustomer (3,600 rows) exceeds CustomerRFM (2,047 rows); the gap likely represents registered users without completed purchases.

4. Data Preparation and Modelling

Power Query Transformations

- Column type enforcement: date columns to Date; Boolean flags to True/False; monetary columns to Decimal Number.
- Sort-by-column assignments: MonthName sorted by MonthNumber; YearMonth sorted by YearMonthSort; WeekdayName sorted by WeekdayNumber.
- DimDate marked as date table, required for time-intelligence DAX functions.
- Measures table created: _Measures = {0}, hidden in model, to organise all 30+ measures cleanly.

Star Schema

FactSales sits at the centre. All relationships use single-directional filtering toward the fact table. BusinessInsights is an unrelated reference table used only on Page 5.

From	To	Cardinality	Active
FactSales.OrderDate	DimDate.Date	Many:1	Yes
FactSales.CustomerID	DimCustomer.CustomerID	Many:1	Yes
FactSales.ProductID	DimProduct.ProductID	Many:1	Yes
CustomerRFM.CustomerID	DimCustomer.CustomerID	1:1	Yes

RFM Customer Segmentation

RFM segmentation was pre-computed in Python (ecommerce_sales_analysis.ipynb) and loaded as CustomerRFM.csv. Customers scored 1-5 on Recency, Frequency, and Monetary dimensions.

Segment	Customers	Combined Revenue
Champions	291	$639,323
At Risk	405	$615,185
Loyal Customers	475	$433,065
Recent Customers	300	$69,462
Needs Attention	246	$66,966
Lost	290	$41,447
Potential Loyalists	40	$23,508

5. Analytical Methodology: DAX Measures

All 30+ measures are stored in dax/DAX_Measures.dax. Six functional groups:

Group	Example Measures	Key Logic
Executive KPIs	Total Revenue, Total Profit, Profit Margin %, Total Orders, AOV	All include IsCompleted = TRUE() internally
Time-Series KPIs	MoM Revenue Change, MoM Growth %, YTD Revenue, Rolling 3M Avg	Require active FactSales to DimDate relationship
Product and Category	Sales Contribution %, Product Revenue Rank	Use ALL(DimProduct) to compute share-of-total
Customer KPIs	Repeat Customers, Repeat Customer Rate %	ADDCOLUMNS + FILTER iterates CustomerID in context
Discount KPIs	Margin on Discounted/Non-Discounted Orders, Discount Impact	Filter by HasDiscount flag
Order Status	Returned Orders, Returned Revenue, Return Rate %, Cancelled Orders	Look across all order statuses intentionally

Validation Targets (from DAX_Measures_Guide.md)

Measure	Expected Value	Source
Total Revenue	$1,888,955.83	DAX_Measures_Guide.md
Total Orders	8,456	DAX_Measures_Guide.md
Total Profit	$929,083.36	DAX_Measures_Guide.md
Profit Margin %	49.2%	DAX_Measures_Guide.md
Total Customers	2,047	DAX_Measures_Guide.md
Total Units Sold	17,840	DAX_Measures_Guide.md
Average Order Value	$223.39	DAX_Measures_Guide.md
Repeat Customer Rate %	64.4%	DAX_Measures_Guide.md

6. Dashboard Walkthrough

Page 1: Executive Overview

Business question: How is the business performing overall?

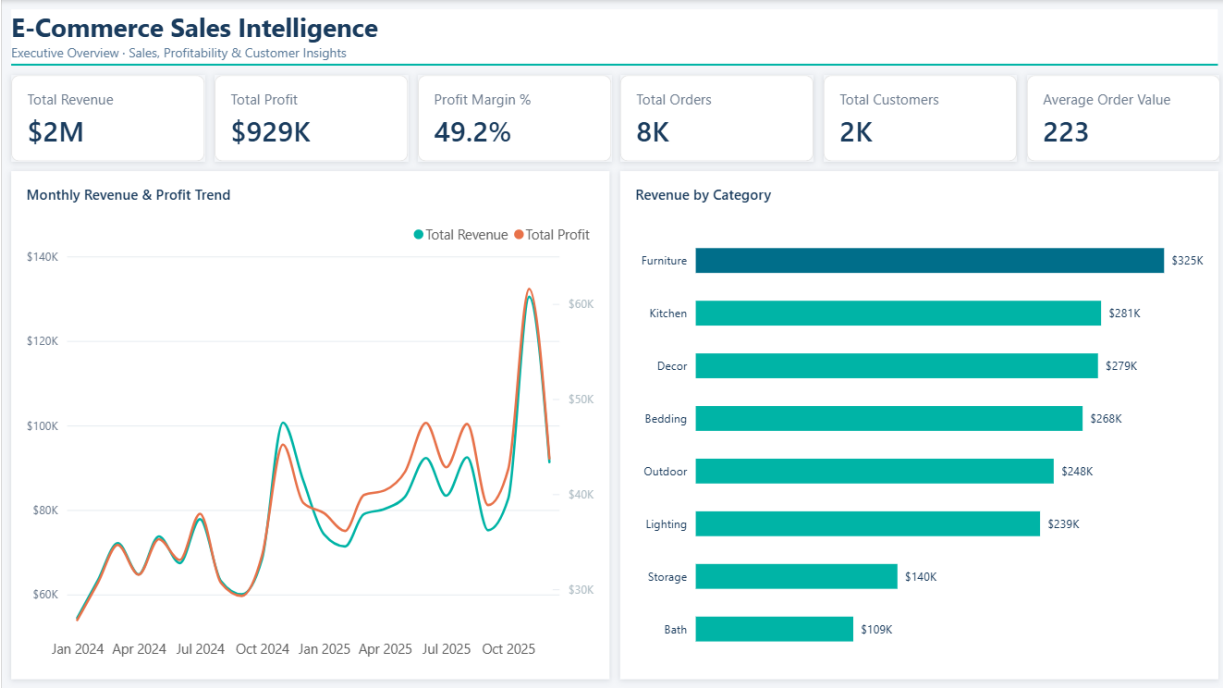


Figure 1: Executive Overview — 6 KPI cards, monthly revenue and profit trend, revenue by category.

- KPI Cards: Total Revenue ($2M), Total Profit ($929K), Profit Margin % (49.2%), Total Orders (8K), Total Customers (2K), Average Order Value ($223).
- Monthly Revenue and Profit Trend: Jan 2024 to Dec 2025. Notable revenue spike in Oct-Nov 2025, consistent with seasonal home-goods demand.
- Revenue by Category: Furniture leads at $325K, followed by Kitchen ($281K), Decor ($279K), Bedding ($268K). Storage ($140K) and Bath ($109K) are the smallest.
- Business use: One-minute leadership health check. Filter to current year for YTD tracking.

Page 2: Sales and Product Performance

Business question: Which products and categories are most profitable?

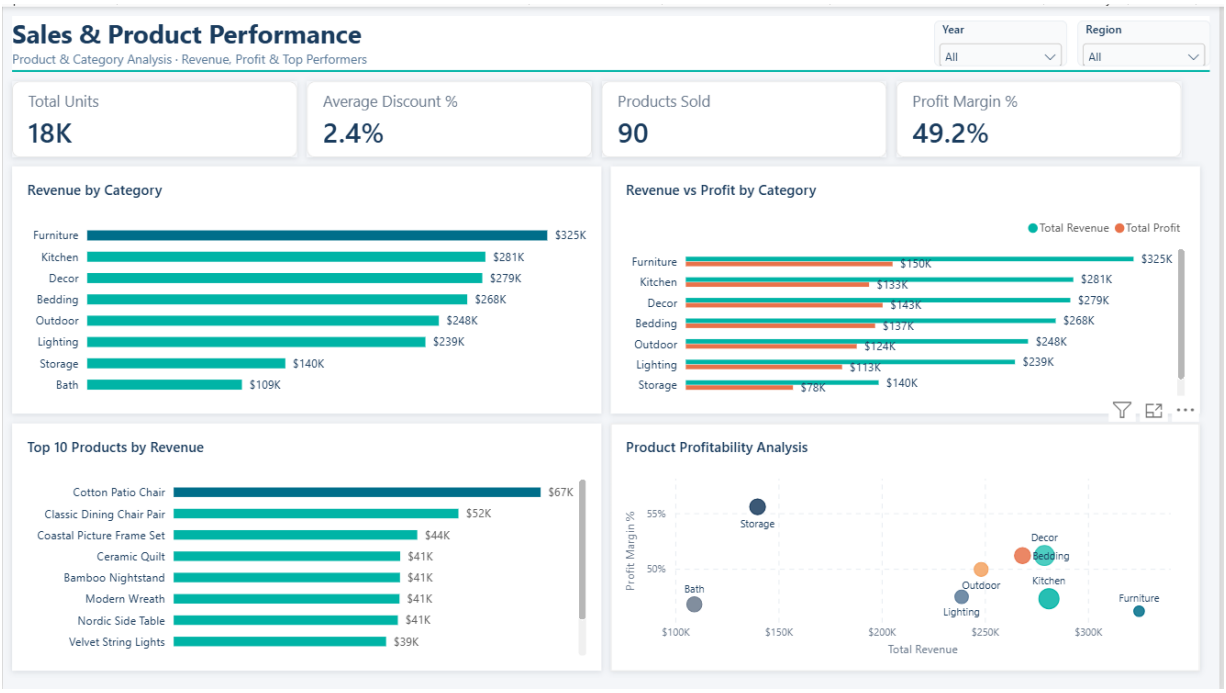


Figure 2: Sales and Product Performance — category comparisons, product profitability scatter, top 10 products.

- KPI Cards: Total Units (18K), Average Discount % (2.4%), Products Sold (90 SKUs), Profit Margin % (49.2%).
- Revenue vs Profit by Category: Furniture generates the highest revenue ($325K) with a proportionally narrower profit bar versus Storage.
- Product Profitability Scatter: Storage sits upper-left (high margin, moderate revenue). Furniture is lower-right (high revenue, lower relative margin).
- Top 10 Products by Revenue: Cotton Patio Chair ($67K), Classic Dining Chair Pair ($52K), Coastal Picture Frame Set ($44K).
- Business use: Category managers use this page to prioritise assortment and pricing decisions.

Page 3: Customer Insights and RFM Segmentation

Business question: Who are the most valuable customers, and how are they segmented?

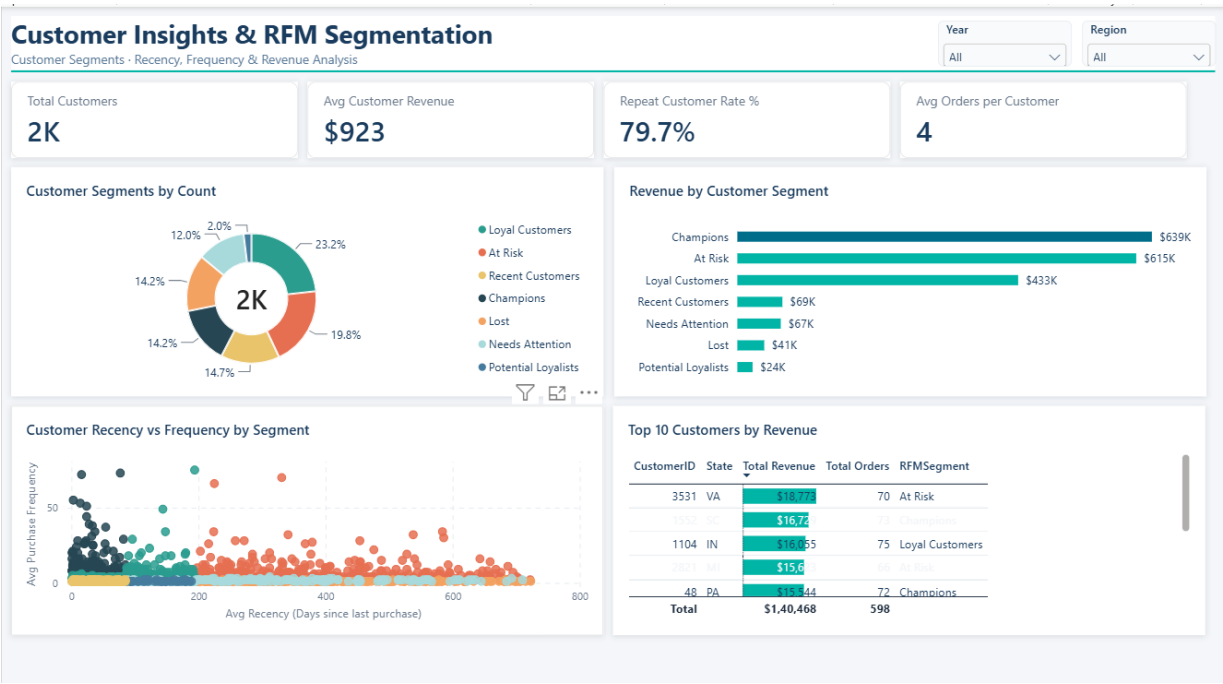


Figure 3: Customer Insights and RFM Segmentation — segment distribution, revenue by segment, recency-frequency scatter.

- KPI Cards: Total Customers (2K), Avg Customer Revenue ($923), Repeat Customer Rate %, Avg Orders per Customer (4).
- Customer Segments Donut: Loyal Customers is the largest segment (23.2%), followed by At Risk (19.8%), Champions (14.2%).
- Revenue by Segment: Champions ($639K) and At Risk ($615K) are the two highest-revenue segments.
- Top 10 Customers Table: Customer 3531 (VA, At Risk) leads at $18,773 with 70 orders.
- Business use: Marketing team targets re-engagement campaigns based on segment filters.

Page 4: Geographic and Channel Performance

Business question: Where do customers come from, and which channels acquire the best customers?

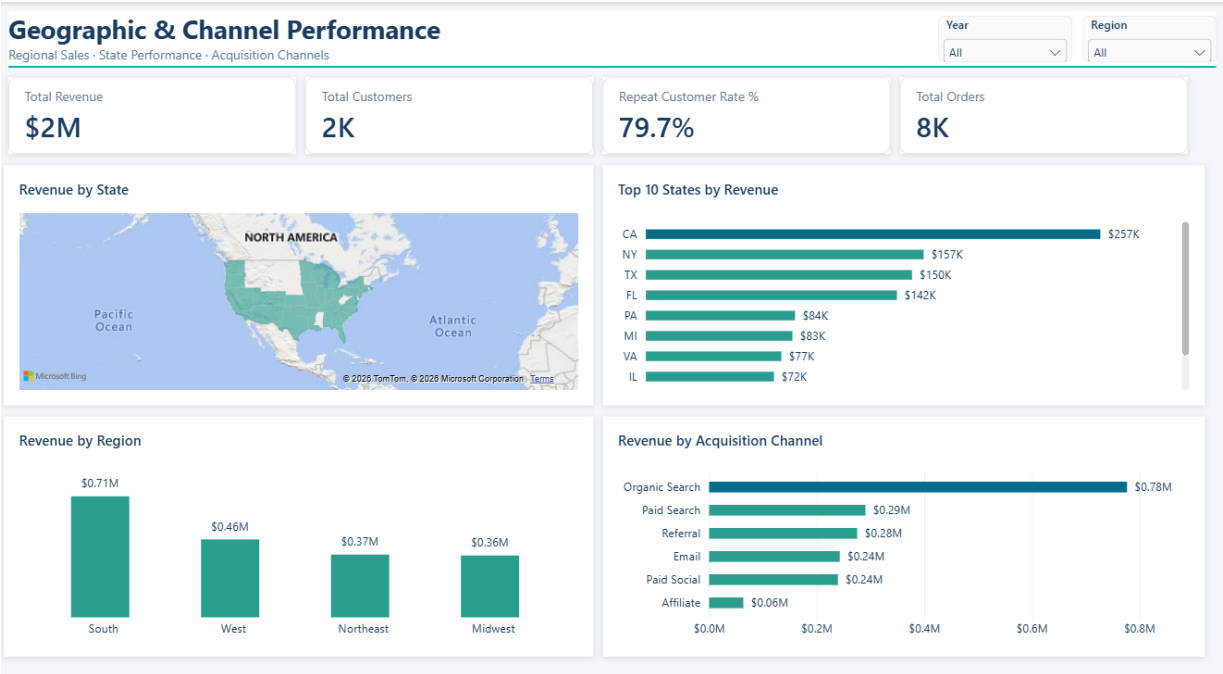


Figure 4: Geographic and Channel Performance — US state map, top 10 states, regional breakdown, channel revenue.

- Revenue by State Map: California, New York, Texas, and Florida are the strongest markets.
- Top 10 States: CA $257K, NY $157K, TX $150K, FL $142K. Top 4 account for ~37% of total revenue.
- Revenue by Region: South dominates ($0.71M), followed by West ($0.46M), Northeast ($0.37M), Midwest ($0.36M).
- Revenue by Acquisition Channel: Organic Search leads at $0.78M, more than double Paid Search ($0.29M).
- Business use: Growth and media teams allocate geographic and channel budgets using this page.

Page 5: Business Insights and Recommendations

Business question: What should the business act on?

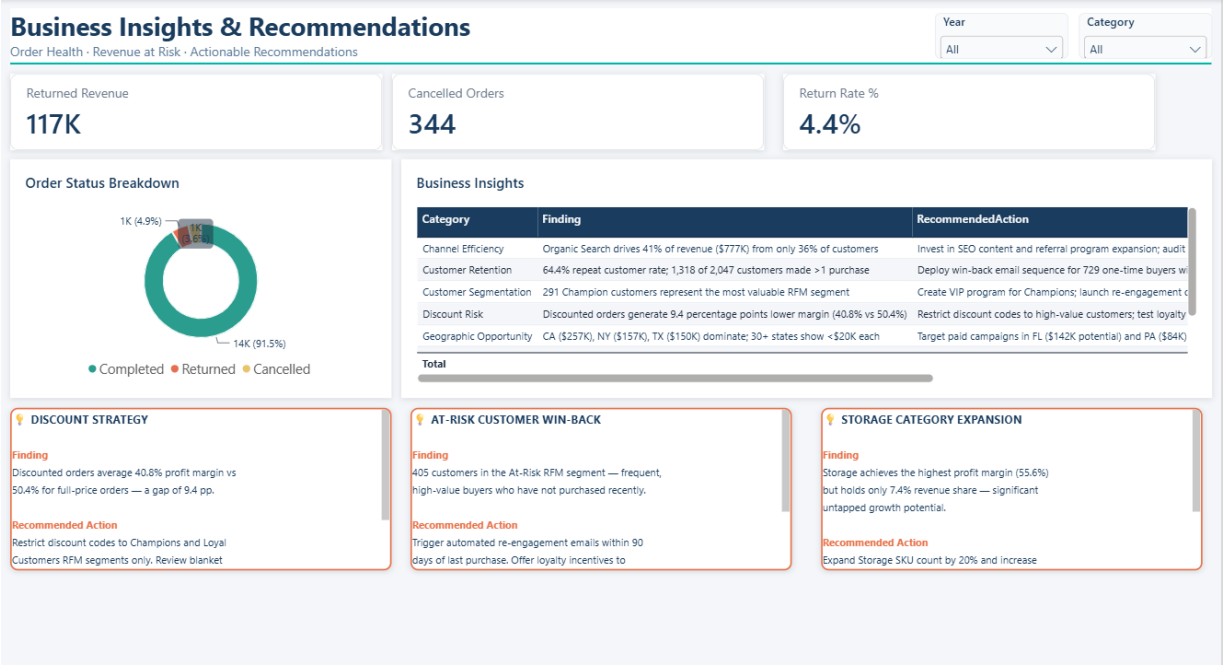


Figure 5: Business Insights and Recommendations — order health KPIs, insights table, three priority action cards.

- KPI Cards: Returned Revenue ($117K), Cancelled Orders (344), Return Rate % (4.4%).
- Order Status Donut: 91.5% completed, 4.9% returned, 3.6% cancelled.
- Business Insights Table: 8 records from BusinessInsights.csv with Category, Finding, and Recommended Action.
- Discount Strategy card: 40.8% margin on discounted orders vs 50.4% full-price (-9.4 pp). Restrict to Champions and Loyal Customers only.
- At-Risk Win-Back card: 405 At-Risk customers ($615K combined past value). Re-engage within 90 days of last purchase.
- Storage Expansion card: Storage achieves highest gross margin (54.1%) but only 7.4% revenue share.

7. Key Findings and Recommendations

All findings are grounded in verified source data. Figures in parentheses indicate the data source.

Finding 1 — Strong Overall Profitability

Evidence: $1,888,956 total completed revenue at 49.2% gross profit margin (FactSales.csv). Month-level peaks reach ~$130K in Oct-Nov 2025, confirming seasonal demand.

Recommendation: Monitor margin quarterly for category mix shifts. Note that this is gross margin before operating costs.

Finding 2 — Discount Margin Erosion

Evidence: Discounted orders: 40.8% margin. Non-discounted orders: 50.4% margin. Gap = 9.6 percentage points (FactSales.csv). Approximately 1,290 discounted orders recorded.

Recommendation: Restrict discount access to Champions and Loyal Customers. Test loyalty tier incentives as non-margin-dilutive alternatives.

Finding 3 — At-Risk Customers Hold Significant Revenue

Evidence: 405 At-Risk RFM customers generated $615,184 in past revenue (CustomerRFM.csv), comparable to Champions ($639K from 291 customers).

Recommendation: Deploy re-engagement email sequence within 90 days of each customer's last purchase, prioritising top-revenue At-Risk customers first.

Finding 4 — Storage Category Is Underweighted

Evidence: Storage achieves the highest average gross margin (54.1% per DimProduct.csv) but generates only $140K in revenue (7.4% of total).

Recommendation: Expand Storage SKU count and allocate incremental marketing budget to this category.

Finding 5 — Geographic Concentration

Evidence: Top 4 states (CA, NY, TX, FL) account for ~37% of total revenue. Over 30 states individually contribute less than $20K (BusinessInsights.csv BI-006).

Recommendation: Shift paid campaign budget toward FL ($142K potential) and PA ($84K), where existing customer bases validate product-market fit.

Finding 6 — Organic Search Efficiency

Evidence: Organic Search generates $778K from 741 customers (revenue per customer ~$1,050). Paid Social and Email each deliver ~$240K (FactSales.csv).

Recommendation: Increase SEO content investment. Audit Affiliate partner mix for quality before scaling that channel.

8. Limitations and Assumptions

- **Gross margin only.** Profit figures are gross profit (Revenue minus COGS). Marketing spend, logistics, CAC, and overheads are excluded. The 49.2% margin is gross, not net.
- **No sales targets.** No SalesTarget table was provided. No variance-to-budget analysis is possible. The PowerBI_Model_Guide.md describes how to add a target table if one becomes available.
- **PBIX internal model.** The PBIX file is confirmed as a valid Power BI package. Page names were confirmed from internal JSON. The .dax file is the authoritative measure source.
- **Repeat Customer Rate discrepancy.** The DAX validation table specifies 64.4%. Pages 3 and 4 show 79.7%. The discrepancy reflects a different denominator in the visual filter context. 64.4% is used in this report as the independently validated figure.
- **RFM is relative.** Segment labels are based on the two-year purchase window. At Risk does not mean certain churn; it means a customer's pattern has diverged from their historical norm.
- **No 2023 data.** YoY measures return BLANK for all of 2024. This is expected behaviour documented in DAX_Measures_Guide.md.
- **Dataset provenance.** The data was created for this portfolio project and contains no real personal or commercial data.

9. Conclusion and Future Enhancements

What the Project Demonstrates

This project demonstrates the ability to design and implement a star schema data model from raw CSV files; author 30+ DAX measures across six functional groups with correct filter context handling; segment customers using an RFM model computed in Python and integrated into Power BI; and build a cohesive five-page dashboard that tells a complete business story from headline KPIs to executive recommendations. The project is fully documented and reproducible from its source files.

Suggested Future Enhancements

Enhancement	Description
Sales target integration	Add SalesTarget table at Month x Category grain; use USERELATIONSHIP for target DAX measures
Forecasting	Extend DimDate to 2026-2027; apply Power BI built-in forecast or Azure ML model output
Customer LTV modelling	Replace simple RFM Monetary with a BG/NBD or Pareto/NBD lifetime value model
Multi-touch attribution	Replace first-touch channel attribution with a multi-touch model using order sequences
Drill-through pages	Add Category detail and Customer profile pages activated from main visuals
Row-level security	Restrict regional managers to their own region data in Power BI Service
Live data refresh	Connect to Azure SQL or Snowflake with scheduled Power BI Service refresh

10. Technical Stack and Reproducibility

Tool	Purpose
Microsoft Power BI Desktop	Dashboard development, DAX measures, data modelling
Python (pandas, NumPy)	EDA, data cleaning, RFM segmentation
Jupyter Notebook	Exploratory analysis and pre-processing
CSV (UTF-8)	All data files
Git / GitHub	Version control and portfolio publication

How to Open and Explore the Dashboard

- Install Microsoft Power BI Desktop (free) from powerbi.microsoft.com.
- Clone or download this repository.
- Open powerbi/Ecommerce_Sales_Intelligence.pbix — data is embedded, no connection needed.
- Navigate the five pages using the tabs at the bottom.
- Use Year, Region, and Category slicers to explore interactively.

To rebuild from scratch: follow docs/PowerBI_Model_Guide.md, then add measures from dax/DAX_Measures.dax using dax/DAX_Measures_Guide.md as the validation reference.
